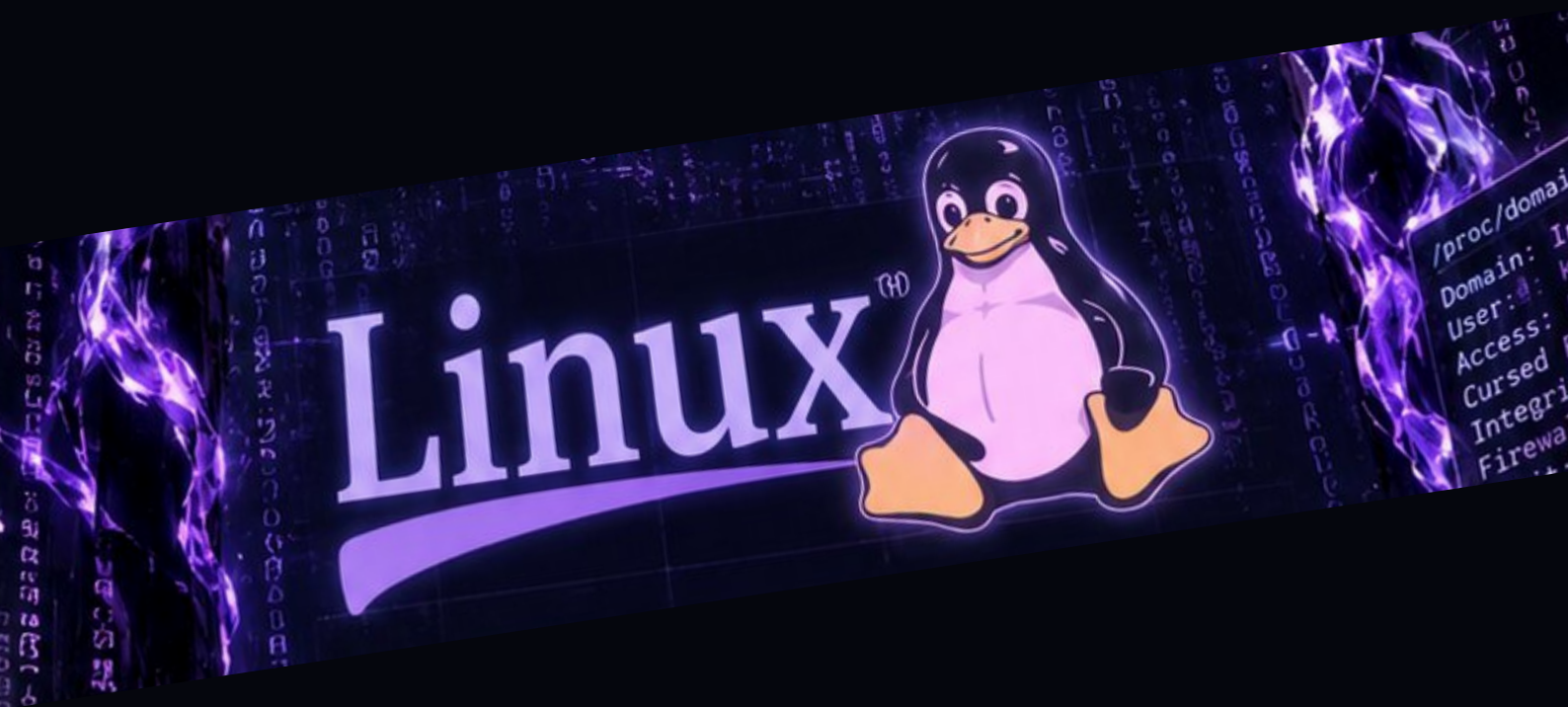


LINUX

TROUBLESHOOTING

DEEP DIVE

Beginner-Friendly Guide for System Administration and Networking



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Linux troubleshooting is not about memorizing commands.

It is about understanding:

- how Linux thinks
- where services communicate
- how logs tell stories
- how networking behaves
- how permissions affect everything
- how small configuration mistakes create huge problems

A good Linux administrator does not panic when something breaks.

They investigate.

This guide is structured like a real-world troubleshooting handbook. Each issue contains:

- symptoms
- possible causes
- investigation steps
- explanation
- solution
- prevention tips

1. HOSTNAME PROBLEMS

Problem: Hostname Does Not Change

You run:

```
hostnamectl set-hostname srv-web
```

But after reboot, the old hostname still appears.

Understanding the Problem

Linux stores hostname information in multiple places:

- runtime hostname
- **/etc/hostname**
- **/etc/hosts**
- cloud-init configurations
- DHCP provided hostname

Sometimes one component overwrites another.

Investigation

Check current hostname

```
hostnamectl
```

Check static hostname file

```
cat /etc/hostname
```

Check hosts file

```
cat /etc/hosts
```

You may find:

```
127.0.1.1 old-hostname
```

This can cause inconsistencies.

Solution

Set hostname correctly

```
sudo hostnamectl set-hostname srv-web
```

Edit hosts file

```
sudo nano /etc/hosts
```

Replace:

```
127.0.1.1 old-hostname
```

With:

```
127.0.1.1 srv-web
```

If Cloud-Init Keeps Overwriting

Cloud servers often use cloud-init.

Check:

```
sudo nano /etc/cloud/cloud.cfg
```

Find:

```
preserve_hostname: false
```

Change to:

```
preserve_hostname: true
```

Why This Happens

Linux boot process includes multiple services.

Some services:

- read hostname
- rewrite hostname
- sync network configuration

This is why troubleshooting is about understanding interactions, not only commands.

2. SSH CONNECTION REFUSED

Symptoms

You try:

```
ssh user@server
```

And get:

```
Connection refused
```

What It Means

The server is reachable.

But no process is accepting connections on port 22.

This usually means:

- SSH service stopped
- firewall blocking
- SSH listening on another port

Investigation Process

Check SSH service

```
sudo systemctl status ssh
```

or on RHEL:

```
sudo systemctl status sshd
```

Check listening ports

```
ss -tulnp
```

Look for:

```
LISTEN 0 128 0.0.0.0:22
```

Check firewall

Ubuntu:

```
sudo ufw status
```

RHEL:

```
sudo firewall-cmd --list-all
```

Check SSH configuration

```
sudo nano /etc/ssh/sshd_config
```

Important lines:

```
Port 22
```

```
PermitRootLogin no
```

```
PasswordAuthentication yes
```

Restart SSH

```
sudo systemctl restart ssh
```

Deep Dive Understanding

SSH problems are usually:

- service problems
- port problems
- firewall problems
- authentication problems

The trick is identifying which layer fails.

Think in layers:

1. Network reachable?
2. Port open?
3. Service running?
4. Authentication working?

3. SERVICE FAILS TO START

Symptoms

```
systemctl start nginx
```

Fails silently or gives:

```
Job for nginx.service failed
```

First Rule of Linux Troubleshooting

Never guess.

Read logs.

Investigation

Check service status

```
sudo systemctl status nginx
```

You may see:

- permission errors
- missing files
- syntax issues
- port conflicts

Read detailed logs

```
journalctl -u nginx
```

For recent logs:

```
journalctl -u nginx -n 50
```

Common Real Cause: Port Already Used

Check:

```
ss -tulnp | grep :80
```

You may discover Apache already uses port 80.

Solution

Stop conflicting service:

```
sudo systemctl stop apache2
```

Or modify configuration.

Beginner Mistake

Many beginners restart services repeatedly without reading logs.

Logs are the truth.

4. DISK SPACE FULL

Symptoms

- server becomes slow
- applications crash
- logs stop writing
- updates fail

Investigation

Check disk usage

```
df -h
```

Find large directories

```
du -sh /*
```

More detailed:

```
du -sh /var/*
```

Common Culprit: Logs

Linux servers can accumulate huge logs.

Check:

```
du -sh /var/log/*
```

Clean Journal Logs

```
sudo journalctl --vacuum-time=7d
```

This removes logs older than 7 days.

Find Huge Files

```
find / -type f -size +500M
```

Deep Dive Understanding

Linux rarely crashes because of CPU first.

Storage problems often create:

- database corruption
- failed boots
- package failures
- authentication issues

Disk monitoring is critical system administration.

5. DNS RESOLUTION PROBLEMS

Symptoms

You can ping IPs:

```
ping 8.8.8.8
```

But domains fail:

```
ping google.com
```

Meaning

Internet connectivity works.

DNS resolution fails.

Investigation

Check DNS configuration

```
cat /etc/resolv.conf
```

You should see:

```
nameserver 8.8.8.8
```

Test DNS

```
dig google.com
```

or:

```
nslookup google.com
```

NetworkManager Systems

Modern Linux may overwrite **resolv.conf**.

Check:

```
nmcli device show
```

Temporary Fix

```
sudo nano /etc/resolv.conf
```

Add:

```
nameserver 1.1.1.1
```

Deep Dive

DNS issues are among the most misleading Linux problems.

Users say:

- internet broken
- server offline
- application dead

But the real issue is name resolution.

Always separate:

- connectivity
- resolution

6. PERMISSION DENIED ERRORS

Symptoms

```
Permission denied
```

This is one of the most common Linux problems.

UNDERSTANDING LINUX PERMISSIONS

Linux permissions involve:

- owner
- group
- others

Displayed like:

```
-rwxr-xr--
```

Investigation

Check file permissions

```
ls -l file.txt
```

Check ownership

```
ls -l
```

Example:

```
-rw-r--r-- root root
```

Meaning:

- owner = root
- group = root

Solution

Change owner

```
sudo chown user:user file.txt
```

Change permissions

```
chmod 644 file.txt
```

Dangerous Beginner Mistake

Never use:

```
chmod 777
```

on everything.

This creates major security risks.

7. LINUX SERVER CANNOT REACH INTERNET

Investigation Method

Think step by step.

Step 1: Check Interface

```
ip a
```

Look for:

- interface UP
- valid IP

Step 2: Check Gateway

```
ip route
```

You should see:

```
default via 192.168.1.1
```

Step 3: Ping Gateway

```
ping 192.168.1.1
```

If this fails:

- switch issue
- VLAN issue
- NIC issue
- cable issue
- VM network issue

Step 4: Ping External IP

```
ping 8.8.8.8
```

If gateway works but internet fails:

- NAT issue
- firewall issue
- upstream problem

Step 5: Test DNS

```
ping google.com
```

Deep Dive Mindset

Networking troubleshooting is about isolation.

Do not jump randomly between commands.

Follow the packet path logically.

8. HIGH CPU USAGE

Symptoms

Server becomes slow.

Investigation

Check processes

```
top
```

Better:

```
htop
```

Find top CPU consumers

```
ps aux --sort=-%cpu | head
```

Possible Causes

- infinite loops
- bad scripts
- overloaded database
- crypto miners
- runaway containers

Deep Dive

CPU problems are usually symptoms.

The real problem may be:

- memory pressure
- bad application logic
- swap exhaustion
- poor optimization

9. MEMORY PROBLEMS

Check RAM Usage

```
free -h
```

Check Swap

```
swapon --show
```

Detect OOM Killer

Linux kills processes when memory exhausts.

Check:

```
dmesg | grep -i oom
```

Understanding OOM

OOM means:

Out Of Memory.

Kernel sacrifices processes to save system stability.

10. SYSTEMD TROUBLESHOOTING

Modern Linux heavily depends on systemd.

Important Commands

List failed services

```
systemctl --failed
```

View boot logs

```
journalctl -b
```

Follow live logs

```
journalctl -f
```

Deep Dive

Systemd centralizes:

- logs
- services
- boot management
- timers
- dependencies

Understanding systemd is essential for modern Linux administration.

11. NETWORKING DEEP DIVE

Understanding Ports

Ports are communication endpoints.

Examples:

- 22 = SSH
- 80 = HTTP
- 443 = HTTPS

Check Listening Services

```
ss -tulnp
```

This is one of the most important troubleshooting commands.

Understanding States

You may see:

- LISTEN
- ESTABLISHED
- TIME_WAIT

Each state tells a story about connections.

12. FIREWALL TROUBLESHOOTING

UFW

Check status:

```
sudo ufw status verbose
```

Firewalld

```
sudo firewall-cmd --list-all
```

IPTables

```
sudo iptables -L -n -v
```

Deep Dive

Firewalls operate at packet level.

A service can work perfectly internally while being unreachable externally.

13. PACKAGE MANAGER PROBLEMS

APT Locked

Error:

```
Could not get lock
```

Investigation

Check processes:

```
ps aux | grep apt
```

Never Delete Locks Immediately

Many beginners remove lock files recklessly.

This can corrupt package databases.

14. BOOT PROBLEMS

System Stuck During Boot

Access Logs

After recovery:

```
journalctl -b -1
```

Shows previous boot logs.

Common Causes

- filesystem corruption
- broken fstab
- failed mounts
- driver problems

15. FILESYSTEM CORRUPTION

Symptoms

- read-only filesystem
- boot failures
- input/output errors

Check Filesystem

Unmount first if possible:

```
sudo fsck /dev/sda1
```

Important

Never run fsck on mounted critical filesystems carelessly.

TROUBLESHOOTING PHILOSOPHY

Good Linux administrators:

- observe first
- investigate carefully
- read logs
- isolate layers
- avoid panic
- document findings

Bad troubleshooting:

- random commands
- restarting blindly
- deleting files without understanding
- copying commands from internet blindly

MOST IMPORTANT LINUX TROUBLESHOOTING COMMANDS

System Information

```
uname -a
```

```
hostnamectl
```

```
uptime
```

Networking

```
ip a
```

```
ip route
```

```
ss -tulnp
```

```
ping
```

```
traceroute
```

Processes

```
top
```

```
htop
```

```
ps aux
```

Logs

```
journalctl
```

```
dmesg
```

```
tail -f
```

STORAGE

```
df -h
```

```
du -sh
```

```
lsblk
```

FINAL ADVICE

Linux troubleshooting becomes easier when you stop thinking:

"What command fixes this?"

And start thinking:

"What exactly is failing?"

Every Linux issue usually belongs to one of these categories:

- networking
- permissions
- services
- storage
- configuration
- DNS
- resources
- dependencies

The more you understand the Linux ecosystem, the less troubleshooting feels like magic.